Product Requirements Document (PRD)

RouteWise: Junction-Centric Adaptive Traffic Advisory Tool

Team Name:	**Cyber Knight**	**Evaluation Gate:**	**Gate 3 (Form 3 - 20 Marks)**
Target Segment:	Urban Peak-Hour Commuters & Couriers	**Date / Deadline:**	**Oct 24, 2026 | 3:30 PM**
Problem:	Commuters waste time at congested junctions without knowing which nearby route is less crowded.	**Scope Focus:**	**Max 3 Must-Haves (Strict Discipline)**

1. Executive Summary & Scope Commitment

RouteWise is an AI-powered traffic advisory system engineered to resolve junction-level traffic bottlenecks in dense urban corridors (Tiruchirappalli, Coimbatore, Madurai). Unlike legacy navigation platforms that route all drivers along identical high-capacity arteries—inadvertently compounding junction gridlock—RouteWise provides **junction-level micro-telemetry**, **multi-criteria alternative route ranking**, and **plain-English cross-route explainability**. This PRD defines the specifications for Gate 3, strictly committing to a **maximum of 3 Must-Have features** to guarantee a robust, end-to-end working demo by 3:30 PM.

2. MoSCoW Feature Prioritization

Features are prioritized using the strict evaluation test: *'If this fails, is there still a viable working demo?'*

CORE MANDATE: MAXIMUM 3 MUST-HAVES (GATE 3 SCOPE DISCIPLINE)

Feature 1 (F1): Real-Time Junction Alternative Discovery & Live Flow Telemetry
Fetches multi-path candidates around congested junctions with live TomTom speed, free-flow baseline, delay (sec), and distance. Backed by synthetic fallback.

Feature 2 (F2): Multi-Criteria Heuristic Ranking Engine
Evaluates routes holistically using penalty score: Time + (Delay × 1.5) + (Distance × 0.05). Sorts routes into 'Fastest & Smoothest', 'Secondary Alternative', and 'Avoid - Bottleneck Trap'.

Feature 3 (F3): Transparent Cross-Route Comparative Explanation Box & Metric Matrix
Computes pairwise trade-off deltas across all alternative routes. Displays visual badges ('Saves 12m delay'), comparative sentences ('Why this route?'), and side-by-side metric tables.

MUST-HAVE	**F1: Junction Alternative Discovery**	Queries live TomTom routing / fallback simulator for multi-path trajectories and bottleneck delay.	CRITICAL: Without routes & telemetry, no advisory can exist.
MUST-HAVE	**F2: Multi-Criteria Ranking**	Balances travel duration, bottleneck delay multiplier, and distance friction.	CRITICAL: Determines which route is objectively superior.
MUST-HAVE	**F3: Comparative Explanation Box**	Generates plain-English comparison cards, badges, and cross-route delta matrix.	CRITICAL: Directly fulfills user trust & requirement #1.

SHOULD-HAVE	F4: 15-Min Predictive Trend	Time-of-day decay sparkline showing anticipated congestion trend.	NON-BLOCKING: Demo works with current telemetry.
SHOULD-HAVE	F5: Preference Weight Sliders	Interactive sliders allowing user to adjust Time vs Fuel vs Choke-Point weights.	NON-BLOCKING: Default balanced profile suffices.
COULD-HAVE	F6: Multi-Palette Theme Switcher	Accessible Dark / Cyber Knight / Emerald / Amber theme toggles.	BONUS: Enhances visual polish and WOW factor.
COULD-HAVE	F7: Incident & Jam Raster Tiles	Live heatmapped tile layer overlay on Leaflet map.	BONUS: Live visual enrichment.
WON'T-HAVE	Turn-by-Turn Audio / Tolls	Voice navigation, wallet integrations, crowdsourced incident chatting.	OUT OF SCOPE: Explicitly omitted for 3:30 PM deadline.

3. Persona to Feature Traceability Matrix

Persona & Goal	Friction Point	Feature Addressing Friction	Live Demo Validation
Karthik Sundaram (28) Daily 2W Commuter (Office Cutoff: 9 AM)	Trapped at Central Bus Stand bottleneck for 22 mins; legacy maps reroute too late.	**F1 + F2:** Identifies secondary bypass 400m away, prioritizing delay avoidance over raw length.	Shows Route 1 bypasses junction, saving 11 minutes of delay.
Selvi Murugan (34) Quick Commerce Courier (8 orders/hr SLA)	Sudden queue at 11th Cross; doesn't know if alternate route is actually faster.	**F3:** Comparative Explanation Box displays: *'+400m extra saves 14 min queue delay'*.	Side-by-side metric table renders explicit time and fuel trade-offs.

4. Technical Architecture & Data Pipeline

RouteWise utilizes an agile, decoupled architecture optimized for fast evaluation and zero demo failure:

- **Client Presentation Layer (Vite + React 18):** High-performance SPA with modern CSS theming, responsive Leaflet map, multi-route cards, and interactive comparison matrix.
- **API Gateway & Service Orchestrator (Node.js + Express):** Handles `/api/routes` requests, validates coordinates, and orchestrates telemetry retrieval.
- **Telemetry Aggregator:** Integrates with TomTom Live Routing & Traffic Flow APIs with an automatic in-memory fallback simulation engine for 100% offline uptime.
- **Ranking & Comparison Engine (server/src/services/routeRankingService.js):** Computes holistic penalty score, extracts comparative cross-route deltas, and synthesizes natural-language justification summaries.

5. Acceptance Criteria (Given / When / Then)

F1: Discovery	Given commuter enters origin and destination near congested junction corridor	When user clicks 'Search Live Alternatives'	Returns 2–3 polyline routes with live duration, free-flow time, delay, and distance in < 1.5s.
F2: Ranking	Given Route A has 27 min (18 min delay) and Route B has 13 min (1 min delay)	When multi-criteria ranking algorithm executes	Route B is badged as 'Rank 1: Recommended Alternative' with lowest penalty score.
F3: Explainability	Given alternative routes have been ranked and rendered	When commuter reviews Route B card	Displays clear reason: 'Why this route? Route B saves 14 mins by avoiding bottleneck, despite 0.9km extra.'

6. Live Demo Measurable Success Metrics (5-Minute Verification)

Metric ID	Metric Name	Target Threshold	Live Demo Verification Procedure
M1	Decision Latency	≤ **10 seconds**	Observer understands which route is superior within 10s of route display.
M2	Time Savings vs Congested Route	≥ **20% faster**	Recommended alternative proves ≥ 20% travel time savings over bottleneck.
M3	End-to-End Query Latency	≤ **1500 ms**	Measured live in browser network tab from search click to map draw.
M4	Explanation Coverage	**100% of routes**	Every single route displays comparative badges and plain-language justification.

7. Edge Cases & Fault Tolerance

- **Third-Party API Rate Limit / Network Outage:** In-memory deterministic synthetic generator automatically catches failed external calls and returns realistic Tier-2 corridor routes, preventing blank screens.
- **Universal Arterial Congestion:** If all alternatives are saturated, system flags network severity and identifies the alternative with lowest deceleration variance and least severe incident probability.
- **GPS Inaccuracy / Missing Telemetry:** Snapping algorithm binds coordinates to nearest valid intersection node with fallback default origin/destination presets.